

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

8. RF/Microwave Measurements

6 hrs · asked in **22 of 24 papers** · 9 syllabus sub-topics · **188 marks, 9.1% of the archive** · **all theory, no numerical marks** → 165 of those 188 marks are **power measurement**, and the tool is picked by one number: how many milliwatts.

READ THIS FIRST

Chapter 8 is descriptive and it repeats harder than any other chapter: five short-note titles (bolometry, double-channel bolometer, circulating calorimetry, spectrum analyzer, VNA) and three long-answer shapes cover everything asked in 24 papers.

1. **“Choose a proper power measuring tool and explain its working principle”** (72 Ma 7.5 mW, 74 Ma airport radar, 73 Bh, 82 Bh BTS) → one table, §8.2. Say which band the number falls in, name the sensor, then give that sensor’s working principle from §8.3 or §8.5. **The table is the mark.**
2. **Calorimetry** (9 papers) → §8.5. Static vs circulating, one formula each, and they are the same formula with m/t replaced by $v d$.
3. **Bolometer bridge** (9 papers) → §8.3. Barretter vs thermistor, single bridge, its two faults, double bridge that cures them.

Three things worth knowing before the paper starts:

- **Every microwave power meter is a substitution instrument.** It never measures RF. It measures how much **dc or low-frequency** power had to be taken away to keep a bridge balanced, or how much a **temperature** rose. Open any power answer with that sentence.
 - **The three bands are the whole decision:** low < 10 mW, medium 10 mW to 10 W, high > 10 W. A radar is quoted in **peak** power, so convert with the duty cycle before choosing (§8.2).
 - Measurement questions are worded as “describe how X and Y are measured”. X and Y are almost always **VSWR meter** and **some power sensor**, which is two separate answers glued together (72 Ash, 70 Bh).
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8.1 Microwave Measurements Vs Low-Frequency Measurements

How are microwave measurements different from low-frequency measurements? List the major RF/MW measurement parameters 2 3 3

HOT 4/24 71 Bh, 73 Bh, 79 Bh, 81 Bh · always the opener to a longer measurement question

Why voltage and current stop being useful

- At low frequency the circuit is **lumped**: V and I are the same everywhere in a wire, so measure them and compute Z , power factor and phase angle.
- At microwaves the line is **distributed**: V and I are functions of **distance** along the line, so a “the voltage” does not exist. In a waveguide there is no unique voltage at all.
- **Power is the invariant**: on a lossless line the transmitted power is the same at every plane. So measure **power**, and get everything else from ratios of powers and from the standing-wave pattern.
- Probes and leads are a **significant fraction of a wavelength**, so any contact instrument loads and detunes what it measures. Measurement is made by **directional coupling** and by **sampling** the field, not by touching nodes.
- Circuit behaviour is described by **S-parameters**, which are themselves defined from travelling waves rather than terminal V and I (§3.1).
- **Cost dodge used in every teaching laboratory**: rather than process microwave signals directly, the source is **1 kHz square-wave modulated**, the microwave signal is detected by a square-law crystal, and all the amplitude information is then read with an ordinary 1 kHz low-frequency amplifier, the **VSWR meter**.

The measurables and their tools (81 Bh asks exactly this list)

Measurable	Instrument
Power (low)	bolometer (thermistor / barretter) bridge, Schottky diode sensor, thermocouple sensor
Power (medium, high)	calorimeter (static, circulating), calorimetric wattmeter, thermistor behind a calibrated attenuator or directional coupler
VSWR, $ \Gamma $	slotted line + VSWR meter, reflectometer
Impedance, unknown load	slotted line (minima-shift method), VNA, magic tee
Frequency	cavity (absorption) wavemeter, frequency counter, slotted line via λ_g
Spectrum, harmonics	spectrum analyzer
S-parameters, insertion and return loss	network analyzer (VNA)
Attenuation, insertion loss	two directional couplers + VSWR meter
Noise figure	noise-source + Y-factor measurement (§8.9)
Field strength, radiation zones	EMF survey meter with isotropic probe (§8.10)

8.2 Microwave Power: Bands, Average Vs Peak, Choosing the Sensor

Choose a proper power measurement tool for the stated source and explain its working principle 8

10

HOT 4/24 72 Ash, 72 Ma, 73 Bh, 74 Ma · 72 Ma 7.5 mW, 74 Ma airport surveillance radar, 73 Bh “low microwave power”

- **Power** = energy dissipated or stored per unit time. At microwaves it is the **average** power that a sensor responds to:

$$P_{av} = \frac{1}{T} \int_0^T v(t) i(t) dt, \quad P_{av} = P_{peak} \times \text{duty cycle}, \quad \text{duty} = \tau f_r = \tau/T$$

- Unit: $P(\text{dBm}) = 10 \log_{10}[P(\text{mW})/1 \text{ mW}]$, so 0 dBm = 1 mW, 30 dBm = 1 W, −30 dBm = 1 μW .

The decision table: this is the answer to every “choose a tool” question

Band	Range	Sensor, and why	Section
Low	< 10 mW	resistance changes with power: thermistor or barretter bolometer bridge (floor about -20 dBm), Schottky diode detector (down to -70 dBm = 100 pW), thermocouple sensor	§8.3, §8.4
Medium	10 mW to 10 W	thermistor bolometer behind a calibrated attenuator or 20 dB directional coupler ; static or circulating calorimeter	§8.3, §8.5
High	> 10 W	temperature changes with power: calorimeter (static for moderate, circulating / flow for kW), calorimetric wattmeter	§8.5

Measurement procedure, common to every power meter (the “zeroing” marks)

1. Attach a **calibrated sensor** to the line port where the unknown power appears; connect it to the power meter.
2. **Turn the RF off and zero the meter** (“zero setting”): this removes sensor self-heating and ambient drift from the reading.
3. Turn the RF on. The sensor reacts, the meter reads, and the reading is by **substitution**: the dc or low-frequency power displaced, or the temperature rise.
4. Correct for the coupler or attenuator in front of the sensor, and for the **calibration factor** (mismatch plus sensor efficiency).

The three PYQ cases worked (numbers checked in meas.py)

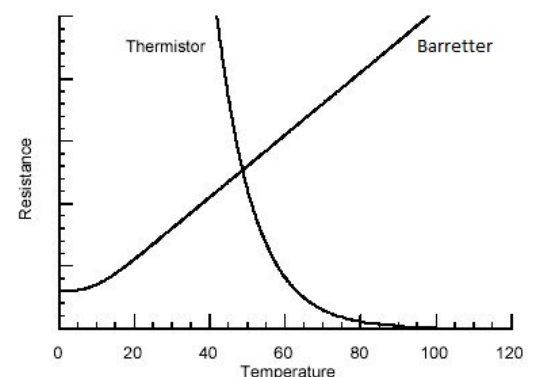
- **72 Ma, “about 7.5 mW”** 10 : 7.5 mW = 8.75 dBm, below 10 mW, so **low power**. Answer: **thermistor bolometer bridge**, sensitivity floor about -20 dBm = 0.01 mW, so 7.5 mW sits comfortably inside its range. Then give the bridge working principle (§8.3). A Schottky diode sensor is the alternative.
- **74 Ma, airport surveillance radar** 8 : a radar is quoted in **peak** power and is pulsed. Take a representative set, 1 MW peak, $\tau = 1 \mu\text{s}$, PRF 1 kHz: duty = 10^{-3} , so $P_{av} = 1$ kW, which is **high power**. Answer: a **circulating (flow) calorimeter** on a dummy load, or a thermistor behind a high-directivity coupler plus a calibrated attenuator; report $P_{peak} = P_{av}/\text{duty}$. Never put a bolometer directly in the line: it would burn out.
- **73 Bh, “low microwave power measurement device”** 8 : name the parameters first (§8.1), then the **bolometer bridge** in full.

8.3 Bolometer Bridge Method

Explain the working principle of bolometry used in microwave measurements 5 6 8 10

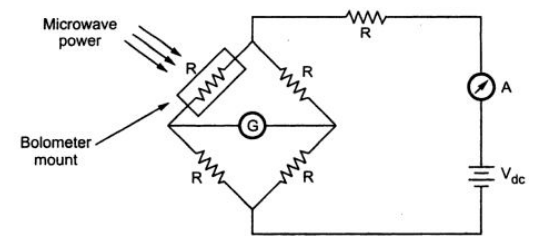
HOT 6/24 70 Bh, 72 Ash, 72 Ma, 73 Bh, 79 Ch, 80 Ch · 79 Ch for 10 marks, 80 Ch as a 5 mark short note

- A **bolometer** is a power sensor whose **resistance changes with temperature** as it absorbs microwave power. RF energy is turned to heat inside the element, the element warms, its resistance moves, and that resistance change is read.
- Bolometer impedances are **100 to 200 Ω** . The **mount** provides a good impedance match, low loss, thermal and mechanical isolation and shielding against leakage.
- **Two types:**
 - **Barretter**: a short thin metal wire (platinum or tungsten, like a fuse), **positive** temperature coefficient. More delicate, so used only for very low power, a few mW at most.
 - **Thermistor**: a semiconductor bead, **negative** temperature coefficient. Small, rugged, easily mounted in a waveguide or coaxial mount; the usual choice. Sensitivity floor about -20 dBm.



Resistance vs temperature: thermistor falls (negative coefficient), barretter rises (positive). Karkee, Ch-8 deck slide 8.

- **Principle:** the element sits in one arm of a **Wheatstone bridge**. With no RF, R_5 is adjusted to balance the bridge. Incident microwave power heats the element, its resistance changes, the bridge unbalances, and the galvanometer or voltmeter current is **calibrated directly in power**.
- **Substitution:** because the element responds only to total heating, the RF power is exactly the **dc power that has to be withdrawn** to restore balance. This is what makes it an absolute measurement rather than a comparison.



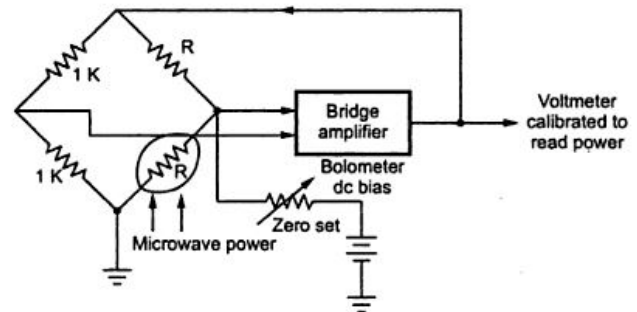
Bolometer mount in one arm of a Wheatstone bridge; the galvanometer current measures the incident power. Deck slide 9.

Single bridge, its two faults, and the double-channel (double bridge) cure 2 4 8

HOT 3/24 80 Ba, 80 Bh, 81 Ba · 80 Ba asks only the limitations 2; 81 Ba is a 4 mark short note

Single bridge

- Balanced under **zero incident power** by adjusting R_5 ; the bolometer arm is R_4 .
- Microwave power into R_4 changes its resistance, unbalances the bridge, and the non-zero output is read on a voltmeter calibrated in power.
- With dc bias voltage E_1 at balance and E_2 after RF is applied, the change $(E_1 - E_2)$ is directly proportional to the microwave power.
- **Two faults** (this is the whole of 80 Ba's 2 marks):
 1. A **mismatch** at the microwave input reflects power, so the element absorbs less than is incident: the reading is wrong and there is no way to tell.
 2. The thermistor is **sensitive to ambient temperature**: room drift is read as RF power. A false reading with no warning.



Self-balancing single bolometer bridge. Deck slide 10.

Double-channel bolometer bridge

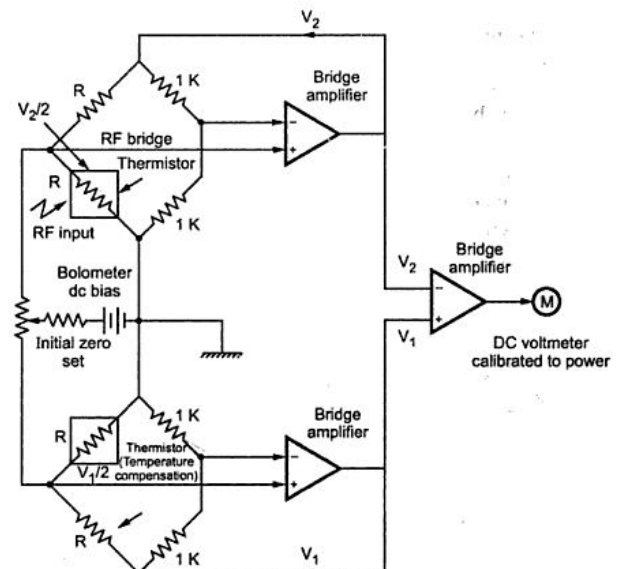
- **Two identical bridges.** The **upper** one carries the RF sensing thermistor and measures the power. The **lower** one carries a matched thermistor with **no RF**, exposed to the same ambient, and compensates temperature drift by forcing $V_1 = V_2$.
- The extra heating caused by mismatch is taken out automatically by **negative dc feedback**: the self-balancing loop lowers the dc power in the sensing thermistor until balance is restored, that is until the **net resistance change is zero**.
- **Zero setting:** with no input, adjust $E_1 = E_2 = E_0$. Without RF the dc voltage across the sensor at balance is $E_1/2$; with RF it is $E_2/2$.
- Average input power is the dc power the RF displaced:

$$P_{av} = \frac{E_1^2}{4R} - \frac{E_2^2}{4R} = \frac{(E_1 - E_2)(E_1 + E_2)}{4R}$$

- If ambient drift moves both rails by ΔE ,

$$P_{av} + \Delta P = \frac{(E_1 - E_2)(E_1 - E_2 + 2\Delta E)}{4R}$$

and since $E_1 + E_2 \gg \Delta E$ in practice, $\Delta P \approx 0$: the meter reads the line above. **That inequality is why the second bridge works.**



Double-channel bolometer bridge: upper bridge measures, lower bridge compensates ambient temperature. Deck slide 12.

- **Worked, to fix the formula** (`meas.py`): $R = 200\ \Omega$, $E_1 = 6\ \text{V}$, $E_2 = 4\ \text{V}$. Sensor dc power falls from 45 mW to 20 mW, so $P_{av} = (36 - 16)/800 = \mathbf{25\ mW}$. A +50 mV ambient drift on both rails would move a **single** bridge to 25.25 mW, an error of 1%, which the second bridge removes.

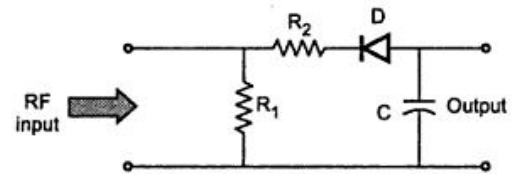
8.4 Schottky Diode Detector and Thermocouple Sensor

Other low-power sensors: the square-law diode detector and the thermocouple syllabus 8.4

no question names them, but both are legitimate answers to “a low power measurement device” (73 Bh, 72 Ash, 72 Ma)

Schottky barrier diode detector

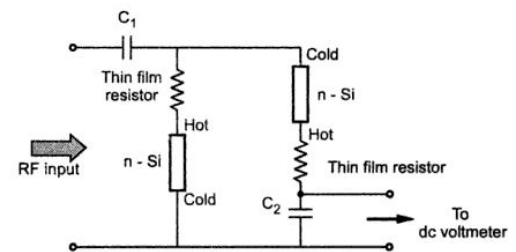
- A zero-biased **metal-semiconductor (point contact) diode** used as a **square-law detector**: output is proportional to the **square** of the input voltage, therefore to input **power**.
- RF input is applied to R_1 and passes through R_2 ; the diode rectifies, and the resulting low-frequency or dc current is read.
- Sensitivity down to **-70 dBm (100 pW)**, up to about 18 GHz. It is the fastest and most sensitive of the low-power sensors.
- The circuit is arranged so the input match does not depend on the diode resistance, which is a strong function of temperature.
- **Caution**: it is square-law only over a limited range. Above roughly -20 dBm it enters a transition region and eventually becomes **linear**, which is exactly why the VSWR meter misreads a high SWR (§8.6).



Schottky barrier diode detector. Deck slide 6.

Thermocouple sensor

- A **thermocouple** is a junction of two dissimilar metals or semiconductors (here n-type Si). A thin film of **tantalum-nitride** resistive load is deposited on a Si substrate and forms one electrode.
- Microwaves are absorbed in the resistive load, heat one end of the junction more than the other, and the junction generates an **emf proportional to incident power**.
- C_2 is the RF bypass capacitor, C_1 the input coupling capacitor (dc block). Emfs of the parallel thermocouples add across C_2 ; the output leads are at RF ground, so the dc voltmeter reads pure dc.
- For a square-wave modulated signal the peak power follows from the average: $P_{peak} = P_{av} T/\tau$, T the period and τ the pulse width. **Same relation as the radar case in §8.2.**
- Wider dynamic range and better match stability than a bolometer, and no dc bias, but no substitution: it must be calibrated.



Thermocouple power sensor. Deck slide 14.

8.5 Calorimetric Methods

What is calorimetry in microwave? Explain how microwave power is measured with a static calorimeter 2 5 6 7 10

HOT 6/24 70 Ma, 71 Ma, 71 Bh, 76 Bh, 77 Ch, 81 Bh · 76 Bh says “static and dry”, which is the same instrument: a dry dielectric load, no flowing water

Calorimetry: the 2 mark definition

- **Calorimetry** measures microwave power by **converting it entirely into heat**, absorbing that heat in a known medium, and measuring the medium’s **temperature rise**. It is the standard method for **medium and high power**, and it is an **absolute** measurement: nothing is compared against a reference source, only against the specific heat of a fluid.
- **Two heating arrangements** (both used with either calorimeter below):
 - **Direct heating**: the rate of heat production is found from the temperature rise of the **dissipating medium itself**.
 - **Indirect heating**: heat is **transferred to another medium** before it is measured.

Static (dry) calorimeter

- A $50\ \Omega$ coaxial line terminated in a lossy dielectric load with a high hysteresis loss, thermally isolated from its surroundings. No fluid circulates, hence “dry”.
- The load dissipates all the microwave power; a thermometer or thermocouple reads the rise T of the thermometric medium over a time t :

$$P = \frac{4.187 m C_p T}{t} \text{ watts}$$

m mass in grams, C_p specific heat in cal/g, T rise in $^{\circ}\text{C}$, t in seconds, and **4.187 is joules per calorie**.

- **Where the formula comes from:** mC_pT is the heat in **calories**, dividing by t makes it calories per second, and 4.187 converts to watts. Quote this and the formula cannot be mis-remembered.
- **Worked** (meas.py): 100 g of water raised 5°C in 60 s is $4.187 \times 100 \times 1 \times 5/60 = \mathbf{34.89\ W}$.
- **Limits:** slow (thermal inertia), and the load must stay isolated, so it suits steady medium power rather than a kilowatt transmitter.

Circulating (flow) calorimeter; circulating Vs flow calorimetry 4 5 8

HOT 4/24 70 Ma, 75 Bh, 81 Bh, 82 Ba · 82 Ba is a 4 mark short note; 81 Bh pairs it with the measurables list of §8.1

- The calorimeter fluid (usually **water**) is **pumped continuously** through a water load in the waveguide. Microwave power heats it in transit, so the **outlet** temperature T_2 exceeds the **inlet** T_1 :

$$P = 4.187 v d C_p (T_2 - T_1) \text{ watts}$$

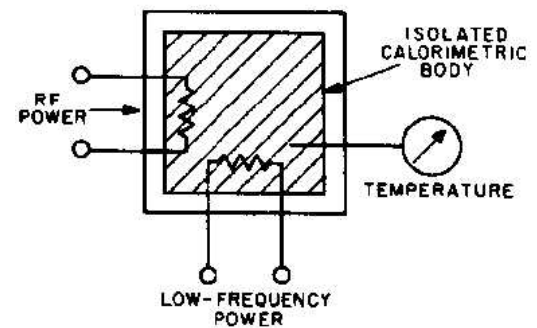
v flow rate in cc/s, d specific gravity in g/cc, C_p specific heat in cal/g.

- **Same law as the static one:** vd is grams per second, which is the static formula's m/t . Confirmed in meas.py.
- **Worked:** 10 cc/s of water with a 2°C rise is **83.7 W**. To take 1 kW with a 10°C rise needs **23.9 cc/s**.
- **Instrumentation:** pump, flow meter, glass tube through the guide, inlet and outlet thermocouples, heat exchanger to return the water cool.

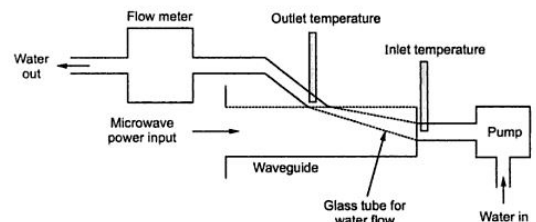
Static Vs circulating (the “differentiate” answer)

	Static (dry)	Circulating (flow)
Load	fixed dry dielectric, thermally isolated	water pumped through continuously
Reading	temperature rise per unit time	steady inlet to outlet difference
Formula	$4.187 m C_p T / t$	$4.187 v d C_p (T_2 - T_1)$
Power handled	medium; the load stores heat and saturates	high , kW upward: the flow carries heat away
Speed	slow, must wait for a measurable rise	continuous once flow is steady
Errors	heat leakage, isolation, thermal inertia	flow-rate stability, thermometer placement, thermal inertia

- **On the wording “circulating Vs flow”** (70 Ma) **verified/added**: the sources use the two words for the **same** instrument. Answer the question that is meant, which is **static vs circulating**, and say so in one line: if a distinction is forced, **circulating** returns the same fluid round a closed loop through a heat exchanger, while a plain **flow** calorimeter passes fluid once and discards it.



Static calorimeter: RF into an isolated calorimetric body, compared against low-frequency power. Deck slide 18.

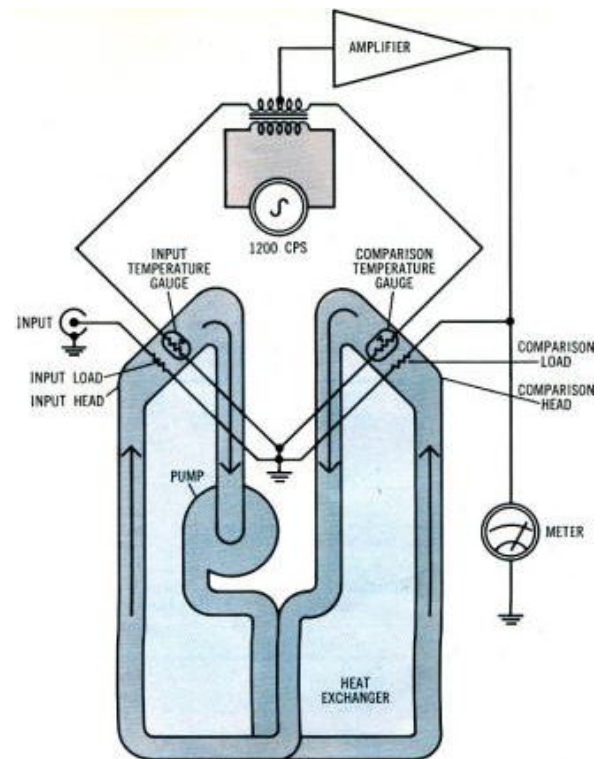


Circulating (flow) calorimeter: water load, pump, flow meter, inlet and outlet thermometers. Deck slide 19.

Power measurement using a calorimeter wattmeter 6

PIN 1/24 80 Ba · paired with the single-bridge limitations of §8.3

- A calorimeter run as a **self-balancing comparison bridge**, so the answer comes out as a meter reading instead of a temperature calculation.
- The unknown RF power is checked against a **1200 cps (Hz) comparison power** in a bridge circuit. Two **temperature-sensitive resistors** serve as gauges.
- **Operation:**
 1. The unknown RF heats an **input load resistor**.
 2. That resistor and one gauge sit in close thermal contact, so the gauge warms and **unbalances the bridge**.
 3. The unbalance signal is **amplified** and applied to the **comparison load resistor**, which is next to the second gauge, and **rebalances** the bridge.
 4. The meter reads the **power supplied to the comparison load**, which at balance equals the unknown RF power.
- Both loads and gauges are **immersed in an oil stream** for efficient, equal heat transfer.
- **Why it is better than a bare calorimeter:** it is a null method, so gauge calibration and ambient drift cancel, and it reads continuously.



Calorimetric wattmeter: input load and comparison load, two temperature gauges, 1200 cps comparison power, oil stream. Deck slide 20.

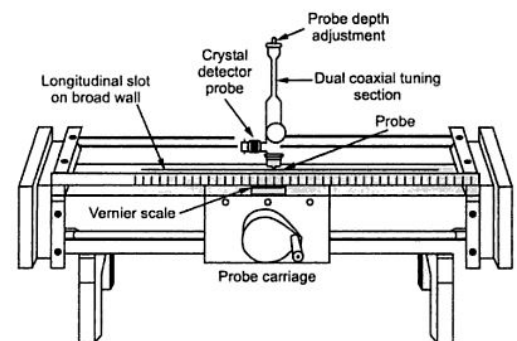
8.6 VSWR Measurement: Slotted Line and VSWR Meter

Describe how standing waves and microwave power are measured with a VSWR meter 2 4

HOT 3/24 70 Bh, 72 Ash, 79 Bh · always part A, with a power sensor as part B: 72 Ash "low power measurement" 8, 70 Bh "bolometry" 6

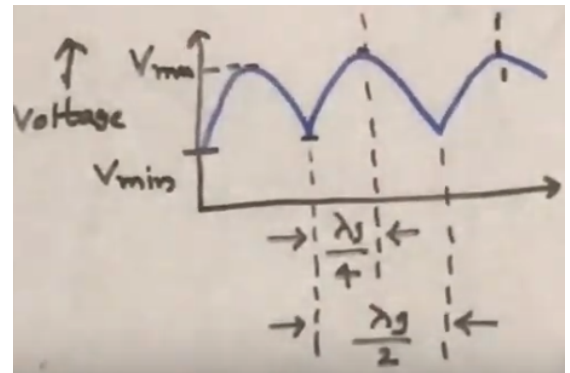
Slotted line carriage: where the standing wave is seen

- A **coaxial E-field probe** penetrates a rectangular waveguide (or coaxial line) through a **longitudinal slot** cut in the centre of the broad wall, over 2 to 3 wavelengths.
- The slot lies where the **wall current has no transverse component**, so cutting it does not disturb the field. Both ends are **tapered** to zero width to reduce discontinuity, and the probe is thin and shallow for the same reason.



Slotted-line carriage: longitudinal slot on the broad wall, probe, crystal detector, vernier scale. Deck slide 22.

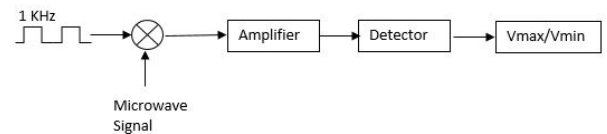
- The carriage moves the probe along the slot at constant depth, so coupling is uniform, and a **vernier scale** reads its position. It carries a tunable stub and a crystal detector, whose 1 kHz output goes to the VSWR meter.
- **It measures:** standing-wave pattern and **VSWR**, **wavelength** ($\lambda_g = 2 \times$ distance between successive minima), and by the minima-shift method **impedance**, **reflection coefficient** and **return loss** (§8.7).



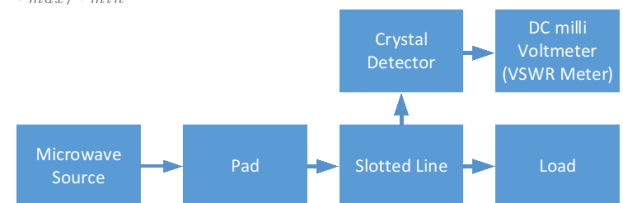
The pattern the probe traces: maxima and minima $\lambda_g/2$ apart, each minimum $\lambda_g/4$ from its neighbouring maximum. Deck slide 27.

VSWR meter

- A **high-gain, high- Q , low-noise voltage amplifier** tuned to a **fixed 1 kHz**, the frequency at which the microwave source is square-wave modulated. Overall gain about 125 dB, switchable in 10 dB steps.
- Input is the **detected** output of the crystal detector; output goes to a **square-law calibrated** voltmeter, which therefore reads V_{max}/V_{min} directly once it has been set to 1.0 at a maximum.
- **Procedure** (low VSWR):
 1. Set the attenuator to about 10 dB, tune the probe stub for maximum reading.
 2. Slide the carriage to a **maximum** and adjust the **gain** until the meter reads **1.0** (0 dB).
 3. Move to the adjacent **minimum**. The meter now reads S **directly**.
- **Three scales:** SWR NORMAL top (1 to 4), SWR NORMAL bottom (3.2 to 10), EXPANDED (1 to 1.3) for a nearly matched load, plus a dB scale. For S between 10 and 40 increase sensitivity by 20 dB, read on the 1 to 4 scale and **multiply by 10**.



1 kHz modulation, mixer, amplifier, detector: the meter reads V_{max}/V_{min} . Deck slide 31.

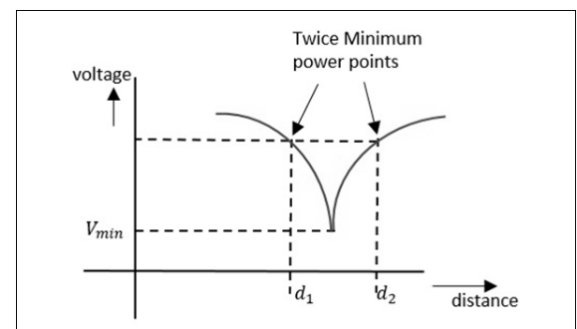


Low-VSWR bench: source, attenuator, slotted line, crystal detector, power meter. Deck slide 28.

How is the VSWR of a microwave transmitter measured when $VSWR > 10$? 6

PIN 1/24 79 Bh · the **double-minimum method**: the one question in the chapter with a derivation in it

- **Why the direct reading fails.** With $S > 10$ the gap between V_{max} and V_{min} is large. To read V_{max} the detector is driven out of its **square-law** region (diode current past about 20 mA, response turning linear), while at V_{min} it is in the noise. A meter calibrated on $I = kV^2$ is then simply wrong, and the error grows with S .
- **The cure:** never look at the maximum. Take **two readings either side of a single minimum**, where the level is low and the detector is safely square-law. This is the **double-minimum method**.
- **Procedure:**
 1. Terminate the line with the load and find a **voltage minimum** on the slotted line; note its position and meter reading.
 2. Move the probe either side until the detected **power is twice the minimum** (voltage $\sqrt{2} V_{min}$). Record the two positions d_1 and d_2 ; $\Delta x = d_2 - d_1$.
 3. Short the line and measure λ_g as **twice** the distance between two successive minima.
 4. Compute S .



The two “twice minimum power” points either side of a minimum, $\Delta x = d_2 - d_1$. Deck slide 29.

- **The formula, and where it comes from.** Near a minimum the detected quantity is $|V|^2 = V_{min}^2 + (V_{max}^2 - V_{min}^2) \sin^2 \beta \delta$ with δ measured from the minimum. Setting $|V|^2 = 2V_{min}^2$ gives $(S^2 - 1) \sin^2 \beta \delta = 1$, so with $\Delta x = 2\delta$ and $\beta = 2\pi/\lambda_g$:

$$S = \sqrt{1 + \frac{1}{\sin^2(\pi \Delta x / \lambda_g)}} \quad \rightarrow \quad S \approx \frac{\lambda_g}{\pi \Delta x} \text{ for large } S$$

- **Both forms verified** in `meas.py` against a simulated standing wave: at $S = 25$ the exact form returns 25.00 and the short form 24.97 (0.12% error); at $S = 60$, 0.04%. **The short form is only for large S :** at $S = 2$ it returns 1.63, 19% out, which is precisely why the direct method is used there instead.
- **Threshold:** the deck sets the changeover at $S > 10$ (which is the wording of 79 Bh), Das at $S > 20$. Say “above about 10, where the detector leaves its square-law region” and the distinction never matters.

8.7 Impedance, Reflection Coefficient and Frequency

Measurement of an unknown load, of reflection coefficient, and of frequency syllabus 8.5–8.8

no direct PYQ, but it is four of the nine syllabus sub-topics, and it is the reasoning behind 74 Bh’s “antenna as DUT” 8 in §7.8

Unknown impedance from the slotted line (minima-shift method)

- Impedance is complex, so **two** numbers are needed: the slotted line supplies S (giving $|\Gamma|$) and d_{min} (giving the phase).
- **Set-up 1:** load connected, measure S and the position of the first minimum. **Set-up 2:** replace the load with a **short**, which puts a minimum at the load plane, and measure the **shift**.
- Shift **towards the generator** means the load is **capacitive**; shift **towards the load** means **inductive**.
- The relations:

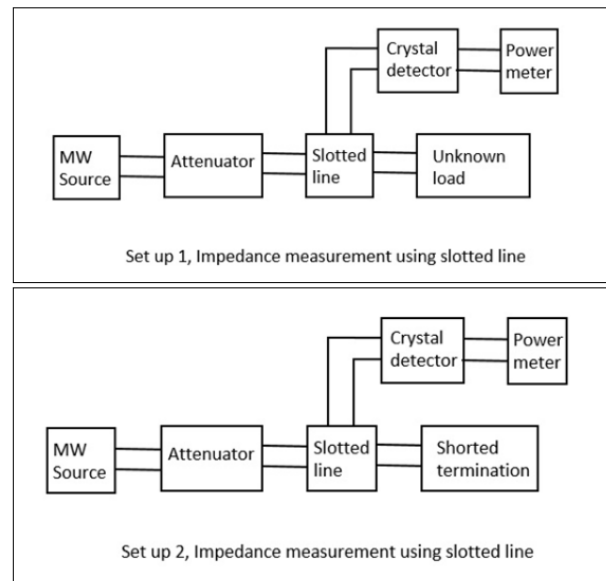
$$\rho = \frac{S-1}{S+1}, \quad \phi_L = 2\beta d_{min} - \pi, \quad \beta = \frac{2\pi}{\lambda_g}, \quad Z_L = Z_0 \frac{1+\Gamma}{1-\Gamma}$$

with $\lambda_g = 2 \times (\text{distance between successive minima})$,
 $\Gamma = \rho e^{j\phi_L}$.

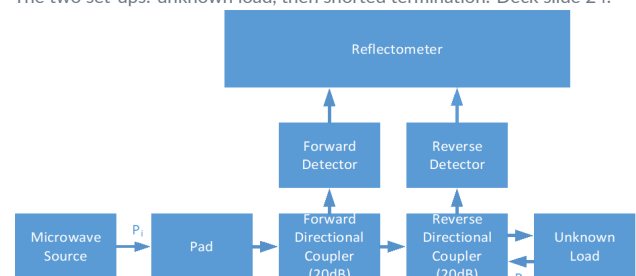
- Equivalent closed form, useful when no chart is at hand,
 $t = \tan \beta d_{min}$:

$$Z_L = Z_0 \frac{1 - jSt}{S - jt}$$

- **Both routes checked** in `meas.py`: a simulated pattern from $Z_L = 60 - j80$ on a 50Ω line is inverted back to $60 - j80$ by each.
- On the Smith chart: draw the S circle, start at the short-circuit point and move **towards the load** by d_{min}/λ_g ; the intersection is z_L .



The two set-ups: unknown load, then shorted termination. Deck slide 24.



Reflectometer: two 20 dB directional couplers sample forward and reverse waves. Deck slide 26.

Reflectometer: $|\Gamma|$ without a slotted line

- Two **directional couplers** face opposite ways: the first samples the **forward** wave at port 4, the second the **reverse** wave at port 3.
- With coupling $C \ll 1$ the ratio of the two detected outputs is the reflection coefficient itself:

$$\left| \frac{b_3}{b_4} \right| = \sqrt{1 - C^2} |\Gamma_L| \approx |\Gamma_L|$$

- Calibrate by shorting port 2 ($|\Gamma| = 1$) and setting the meter to unity, then connect the load. Everything else follows:

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \quad \text{RL(dB)} = -20 \log_{10} |\Gamma_L|$$

- Broadband and fast**, but accuracy is set by coupler **directivity**, so it suits **low VSWR**; a high SWR still wants the slotted line.
- Related loss family, all measured the same way: **insertion loss** = reflection loss + attenuation loss, with reflection loss = $-10 \log(1 - |\Gamma|^2) = -10 \log[4S/(1 + S)^2]$. **Return loss compares P_r with P_i ; insertion loss compares P_0 with P_i (§3.4).**

Frequency measurement

- Cavity (absorption) wavemeter:** a cylindrical cavity with a **variable short**, coupled to the guide through a hole in the narrow wall. Tuning the plunger changes the resonant frequency; at resonance the cavity **absorbs**, and the power meter downstream shows a **dip**. The plunger scale is calibrated in frequency.
 - TE_{011} has the highest Q and no axial current, but it is a higher-order mode, so practical wavemeters use the dominant TM_{010} , excited by magnetic coupling. Polytron absorber behind the plunger stops oscillation above it.
 - Accuracy follows Q : **1% at $Q = 1000$, 0.005% at $Q = 50\,000$.**
- Slotted line:** short the guide, measure λ_g as twice the minima spacing, then

$$\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}, \quad f = \frac{c}{\lambda}, \quad \lambda_c = 2a (TE_{10})$$

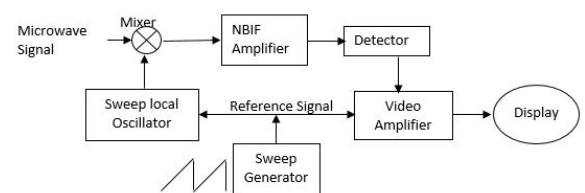
Worked (meas.py): WR-90 ($a = 2.286$ cm) with $\lambda_g = 4.00$ cm gives **9.96 GHz**.

- Frequency counter with a down-conversion front end:** the unknown is mixed with a selected harmonic nf_1 of a known oscillator, and the counter reads $f_x = \text{counter} + nf_1$. Direct and most accurate.

8.8 Spectrum Analyzer**Spectrum analyzer (short note): working principle and block diagram** 4 5

PIN 2/24 80 Bh, 80 Ch · short note both times, so a block diagram plus six lines is the full answer

- A **swept-tuned superheterodyne receiver** that plots **amplitude against frequency**: frequency on the horizontal axis, amplitude on the vertical. It shows what an oscilloscope cannot, namely the **signal spectrum**.
- Working:**
 - The input microwave signal is mixed with a **local oscillator swept electronically** between two limits at a linear rate by a **sawtooth sweep voltage**.
 - The difference product is amplified by a **narrow-bandwidth IF amplifier**, then detected and **video amplified**.
 - The same sawtooth drives the horizontal deflection, with **zero flyback time**, so horizontal position tracks the LO frequency while the detected amplitude gives vertical deflection.
- Measures:** carrier power, harmonics, spurious products, sidebands, phase noise, occupied bandwidth; on a modern analyzer, demodulated WLAN, GSM or LTE signals.
- Four design parameters:** sweep rate, sweep range, IF bandwidth, IF centre frequency.
- Resolution is the IF bandwidth (RBW).** Narrow RBW resolves closer tones but needs a **slower sweep**, because the IF filter must have time to settle: sweep time rises as $1/\text{RBW}^2$. Keep the span as small as the measurement allows.
- Image response:** any signal $f_i = f_0 \pm f_{IF} = f_s \pm 2f_{IF}$ also beats to the IF. Choose f_{IF} **high** to push the image out of band. Worked: $f_{IF} = 450$ kHz puts the image **900 kHz** from the signal; $f_{IF} = 2$ MHz moves it to **4 MHz** away, easily filtered (meas.py).



Block diagram: mixer, swept local oscillator, narrow-band IF amplifier, detector, video amplifier, display. Deck slide 33.

- **Types:** swept-tuned (superheterodyne), FFT, real-time, audio.

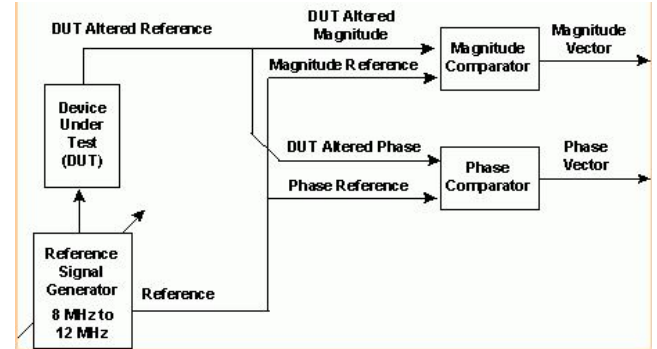
8.9 Network Analyzer, VNA and the Two-Port Model

Describe the working principle of a network analyzer; VNA (short note) 5 8

HOT 3/24 71 Bh, 73 Ma, 78 Ch · 78 Ch as a 5 mark short note on “VNA type network analyzer”; 71 Bh asks the two-port model below

Why it exists, and how it works

- The slotted line measures amplitude and phase **one frequency at a time**, so broadband testing with it is slow and expensive. A **network analyzer** measures both **over a wide band** in a reasonable time.
- **Principle: everything is measured against a reference.**
 1. A **sweep oscillator** feeds a **power divider**, splitting the signal into a **test channel** and a **reference channel**.
 2. The test signal passes through the **DUT**; the reference passes through a **phase-equalising length** of line.
 3. Microwave frequencies cannot be processed directly, so a **harmonic frequency converter** brings both to a fixed IF. A **phase-locked loop** makes the local oscillator track the reference channel, which is what allows a **swept** measurement.
 4. Conversion is in two steps, RF to an MHz IF, then to a kHz IF, where the **magnitude and phase comparators** compare test against reference.
- **Reflection and transmission** are separated by a test unit built from **dual directional couplers**: S_{11} and S_{21} from one direction, then source and load are interchanged for S_{22} and S_{12} . Each S_{ij} is a **ratio of a coupler output to the reference**, with the phase as the difference $\phi_j - \phi_i$.
- **Two types: SNA** (scalar) measures amplitude only; **VNA** (vector) measures **amplitude and phase**, so it can apply **vector error correction** and is far more accurate.



VNA: DUT alters the reference in magnitude (resistive) and phase (reactive); two comparators output the magnitude and phase vectors. Deck slide 36.

	Spectrum an.	Network an.
Measures	unknown signals: power, harmonics, sidebands, noise	known stimulus: Γ , S -parameters, insertion and return loss
Accuracy	lower	higher, vector error correction
IF BW	wider	narrower
Quantity	scalar only	amplitude and phase
Sweep	frequency	frequency and power

Design a two-port network model and derive the required parameters 10

PIN 1/24 71 Bh · the only 10 mark question in §8.9; it wants the set of two-port descriptions and why S is the microwave one

- **The model:** a black box with two ports, port 1 (V_1, I_1) and port 2 (V_2, I_2), both currents taken as flowing **into** the box. Any two of the four are independent, which is what gives the six standard parameter sets.

Set	Defining equations	Element found by	Needs
Z (open-circuit)	$V_1 = Z_{11}I_1 + Z_{12}I_2$ $V_2 = Z_{21}I_1 + Z_{22}I_2$	$Z_{11} = V_1/I_1 _{I_2=0}$	open circuit at a port
Y (short-circuit)	$I_1 = Y_{11}V_1 + Y_{12}V_2$ $I_2 = Y_{21}V_1 + Y_{22}V_2$	$Y_{11} = I_1/V_1 _{V_2=0}$	short circuit at a port
h (hybrid)	$V_1 = h_{11}I_1 + h_{12}V_2$ $I_2 = h_{21}I_1 + h_{22}V_2$	$h_{21} = I_2/I_1 _{V_2=0}$	one short, one open
$ABCD$ (chain)	$V_1 = AV_2 - BI_2$ $I_1 = CV_2 - DI_2$	cascades by matrix product	short and open
S (scattering)	$b_1 = S_{11}a_1 + S_{12}a_2$ $b_2 = S_{21}a_1 + S_{22}a_2$	$S_{11} = b_1/a_1 _{a_2=0}$	matched Z_0 termination

- **Why S at microwaves** (the same argument as §3.1, and the reason this question sits in the measurements chapter): Z , Y , h and $ABCD$ all need a **short or open circuit** at a port. At microwaves a short is not a short (lead inductance), an open radiates, and both are **frequency dependent** and may destabilise an active device. A **matched load** is realisable at every frequency, and a , b are travelling-wave quantities that a directional coupler can sample: that is exactly what the network analyzer above does, which is why S -parameters are the set that can actually be **measured**.

- **Reciprocity** $S_{12} = S_{21}$, **losslessness** $[S]^\dagger [S] = [I]$, **matched** $S_{ii} = 0$; conversions to and from Z exist, $[S] = ([Z] - [I])([Z] + [I])^{-1}$ for Z normalised to Z_0 (§3.2).

Noise figure measurement syllabus 8.9, never asked

- **Y-factor method**: drive the DUT with a calibrated **noise source** that switches between hot and cold. Y is the ratio of the two output powers, and with the source's **excess noise ratio**:

$$F = \frac{\text{ENR}}{Y - 1}, \quad T_e = \frac{T_h - Y T_c}{Y - 1}, \quad F = 1 + \frac{T_e}{T_0}$$

- Worked (meas.py): ENR 15 dB with $Y = 10$ dB gives $F = 3.51$, i.e. **5.46 dB**, equivalently $T_e = 729$ K. The floor any such measurement sits on is $kT_0 = -174$ dBm/Hz.

8.10 Surveying the Radiation Zones Round a BTS

Working principle of a microwave measuring device to map the radiation zones around a GSM BTS 8

PIN 1/24 82 Bh · paired with “how microwaves are hazardous” 4 from §7.2; the zone formulas are §7.1

- **The instrument**: a **broadband EMF survey meter** (field-strength meter), which is a calibrated probe plus a display, usually backed by a **spectrum analyzer** when the individual operators' carriers must be separated.
- **Probe**: an **isotropic (three-axis) probe**, three mutually perpendicular dipoles (E field) or loops (H field) whose outputs are squared and summed, so the reading does not depend on how the probe is held. Diode or thermocouple detection, high-resistance leads so the cable does not perturb the field.
- **Why both E and H must be measured in the near field**: only in the far field is $E/H = 377 \Omega$ and $S = E^2/377$. Closer in, the ratio is not fixed, so the two are measured **separately** and compared against separate limits (§7.1).
- **Working principle**: the probe dipoles develop a voltage proportional to the field; a square-law diode detector gives a dc output proportional to **power density**; high-resistance lines carry it to a meter calibrated in V/m, A/m or W/m².
- **Procedure**: zero the meter with the transmitter off, then walk a **radial grid** out from the tower at a fixed height, log position with GPS, and plot contours. Compare each contour with the ICNIRP or FCC limit for that frequency, and mark the compliance boundary.
- **Worked** (meas.py, same formulas as §7.1): a 1.3 m panel at 1800 MHz has $\lambda = 16.7$ cm, so the reactive near field ends at $0.62\sqrt{D^3/\lambda} = \mathbf{2.25\ m}$ and the far field starts at $2D^2/\lambda = \mathbf{20.3\ m}$. With 20 W into a 17 dBi sector, S at 50 m is 0.032 W/m², against the ICNIRP public limit of $f/200 = 9$ W/m²: a margin of **about 280 times**.
- **Related ask**: 74 Bh's “antenna as a DUT, choose a measurement tool” 8 wants a **VNA** instead (§8.9): return loss and S_{11} give the match and bandwidth, and an anechoic range gives the pattern. It is filed in §7.8.

8.11 PYQ Mapping

Theory questions, covered in this document

- **Microwave vs low-frequency measurements; major measurement parameters** → §8.1. 71 Bh 3, 79 Bh 2, 81 Bh 3
- **Choose a proper power measuring tool and explain it** → §8.2. 72 Ma 10 (7.5 mW), 74 Ma 8 (airport radar), 73 Bh 8 (low power)
- **Bolometry; low power measurement** → §8.3. 70 Bh 6, 72 Ash 8, 79 Ch 10, 80 Ch 5
- **Single-bridge bolometer limitations; double-channel bolometer bridge** → §8.3. 80 Ba 2, 80 Bh 8, 81 Ba 4
- **Calorimetry; static (dry) calorimeter** → §8.5. 70 Ma 2, 71 Ma 5, 71 Bh 7, 76 Bh 10, 77 Ch 6, 81 Bh 6
- **Circulating (flow) calorimetry** → §8.5. 70 Ma 8, 75 Bh 5, 82 Ba 4
- **Calorimeter wattmeter** → §8.5. 80 Ba 6
- **Standing waves and VSWR meter** → §8.6. 70 Bh 4, 72 Ash 2
- **VSWR of a transmitter with VSWR > 10 (double-minimum method)** → §8.6. 79 Bh 6
- **Spectrum analyzer** → §8.8. 80 Bh 4, 80 Ch 5
- **Network analyzer; VNA** → §8.9. 73 Ma 8, 78 Ch 5
- **Two-port network model and its parameters** → §8.9. 71 Bh 10
- **Measuring device to map radiation zones round a GSM BTS** → §8.10. 82 Bh 8

Syllabus topics with no PYQ in 24 papers

- Thermocouples (§8.4), impedance measurement, measurement of unknown loads, measurement of reflection coefficient and frequency measurement (§8.7), and noise (§8.9). All six are written up, briefly, because the “choose a tool” questions can reach any of them.

Numerical content

- **This chapter has no numerical component:** no paper in 24 sets a measurement calculation. The arithmetic above (duty cycle, the two calorimeter laws, the bridge substitution, the double-minimum formula, λ_g to frequency) exists to make the descriptive answers specific, and every number is re-derived in `src/meas.py`.

Corrections and source defects found while building this chapter

- **Karkee's Chapter-8 PDF export is missing seven slides** that the pptx has: impedance measurement, the reflectometer and the low and high VSWR methods (deck slides 23 to 30). The **double-minimum method**, which is the whole of 79 Bh's 6 marks, appears **only** in the pptx. Figures are pulled from the pptx for this reason.
- Deck slide 30, "Measurement of frequency", is an **empty slide**. §8.7 is written from Das ch. 13 instead verified/added.
- Das Fig. 13.21's worked chart example ($S = 2$, $d_{min} = 0.2\lambda_g$) prints $Z_L = Z_0(1.0 + j0.7)$. Its own equations 13.46 to 13.50 give $Z_0(1.55 - j0.69)$, and the closed form $Z_0(1 - jSt)/(S - jt)$ agrees with that. The printed point does sit on the $S = 2$ circle, but at 70.7° , which needs $d_{min} = 0.348\lambda_g$, not 0.200. **The answer does not follow from its own data**; both routes are asserted in `meas.py`.
- Das §13.8 writes insertion loss as $10 \log(P_0/P_i)$, which is negative for a lossy network. Losses are quoted positive: $10 \log(P_i/P_0)$. Same slip as the Chapter-3 deck's (§3.4).
- Deck slide 19 titles the circulating calorimeter "**Circular** Calorimeter"; no such instrument. It is circulating, or flow.
- The deck sets the high-VSWR threshold at $S > 10$, Das at $S > 20$. Both are quoted in §8.6 rather than one being picked.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.